



Natural Language Processing: Advancements in Transformers

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Table of Content

- A theory of positional encodings relativity
- Your time, FFN
- Advanced attention



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“Only the sith deal in absolutes” ©

$$\text{RelativeAttention} = \text{Softmax} \left(\frac{QK^{\top} + S_{rel}}{\sqrt{D_h}} \right) V$$

Rotate Query, Key (or other activations if you know that you're doing)

$$\mathbf{q}_m^\top \mathbf{k}_n = (\mathbf{R}_{\Theta, m}^d \mathbf{W}_q \mathbf{x}_m)^\top (\mathbf{R}_{\Theta, n}^d \mathbf{W}_k \mathbf{x}_n) = \mathbf{x}^\top \mathbf{W}_q \mathbf{R}_{\Theta, n-m}^d \mathbf{W}_k \mathbf{x}_n$$

This is just a rotation matrix for each pair of coordinates

$$\mathbf{R}_{\Theta, m}^d = \begin{pmatrix} \cos m\theta_1 & -\sin m\theta_1 & 0 & 0 & \cdots & 0 & 0 \\ \sin m\theta_1 & \cos m\theta_1 & 0 & 0 & \cdots & 0 & 0 \\ 0 & 0 & \cos m\theta_2 & -\sin m\theta_2 & \cdots & 0 & 0 \\ 0 & 0 & \sin m\theta_2 & \cos m\theta_2 & \cdots & 0 & 0 \\ \vdots & \vdots & \vdots & \vdots & \ddots & \vdots & \vdots \\ 0 & 0 & 0 & 0 & \cdots & \cos m\theta_{d/2} & -\sin m\theta_{d/2} \\ 0 & 0 & 0 & 0 & \cdots & \sin m\theta_{d/2} & \cos m\theta_{d/2} \end{pmatrix}$$

$$\theta_{p,j} = p \cdot \omega_j = p \cdot \frac{1}{\text{base}^{j/d}}$$

Attention with linear biases (ALiBi)

$$\begin{bmatrix} q_1 \cdot k_1 & & & & \\ & q_2 \cdot k_1 & q_2 \cdot k_2 & & \\ & q_3 \cdot k_1 & q_3 \cdot k_2 & q_3 \cdot k_3 & \\ & q_4 \cdot k_1 & q_4 \cdot k_2 & q_4 \cdot k_3 & q_4 \cdot k_4 \\ & q_5 \cdot k_1 & q_5 \cdot k_2 & q_5 \cdot k_3 & q_5 \cdot k_4 & q_5 \cdot k_5 \end{bmatrix} + \begin{bmatrix} 0 & & & & \\ -1 & 0 & & & \\ -2 & -1 & 0 & & \\ -3 & -2 & -1 & 0 & \\ -4 & -3 & -2 & -1 & 0 \end{bmatrix} \cdot m$$

m is a constant with **one value per head**:

$$m_i = \frac{1}{2^i \cdot \text{scale}}, \text{scale}=0.5 \text{ (for example)}$$



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You can also play around with the vanilla FFN

$$\text{FFN}_{\text{GLU}}(x, W, V, W_2) = (\sigma(xW) \otimes xV)W_2$$

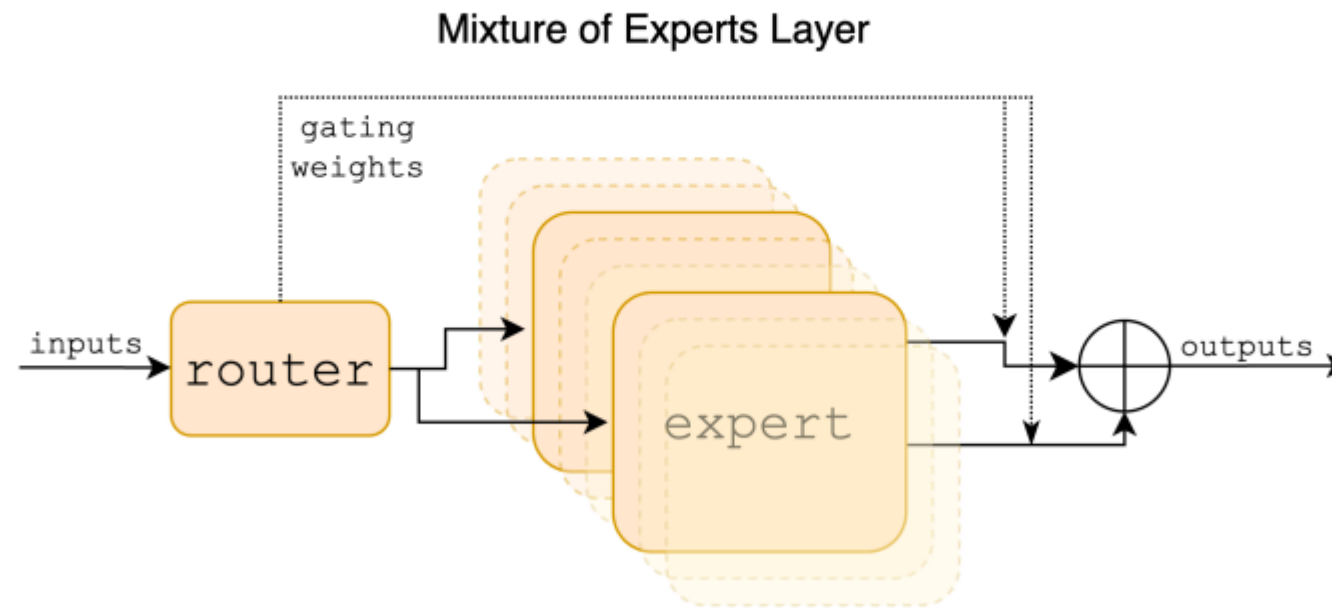
$$\text{FFN}_{\text{Bilinear}}(x, W, V, W_2) = (xW \otimes xV)W_2$$

$$\text{FFN}_{\text{ReLU}}(x, W, V, W_2) = (\max(0, xW) \otimes xV)W_2$$

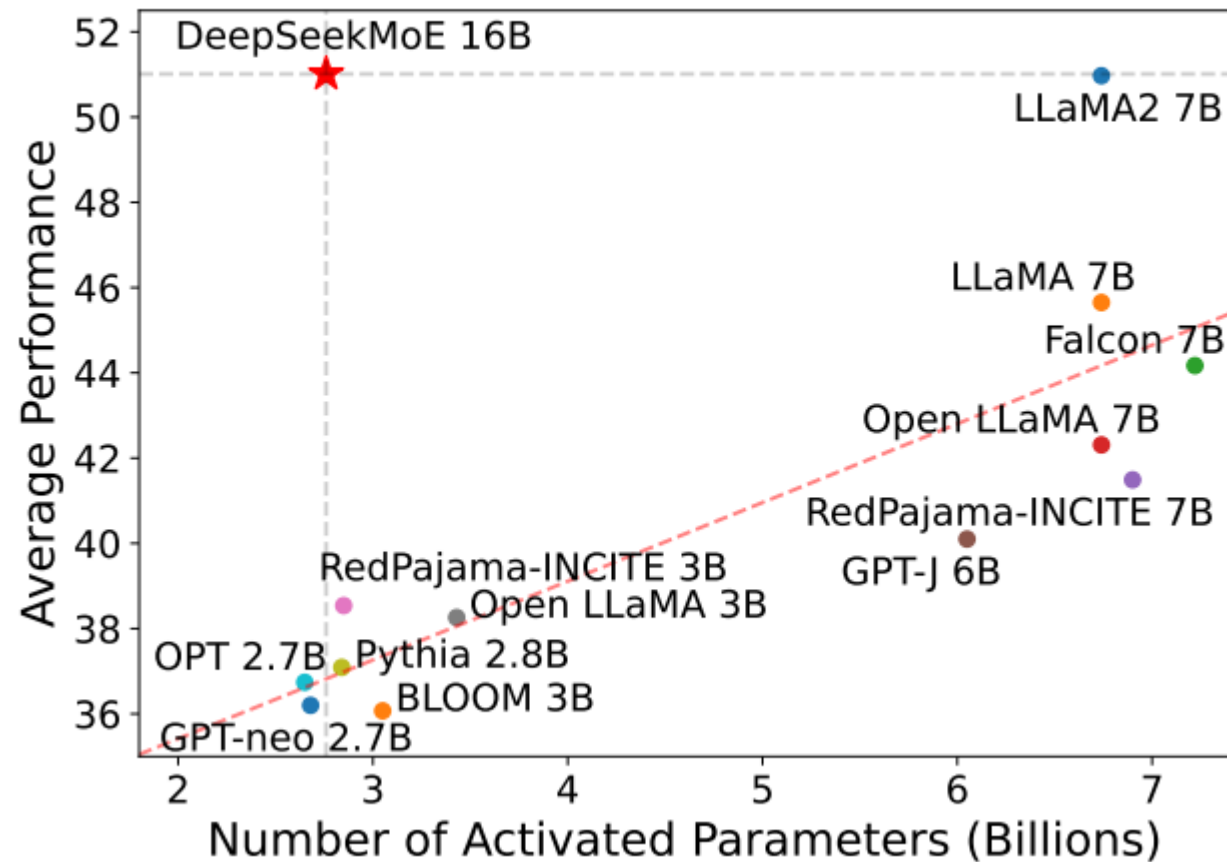
$$\text{FFN}_{\text{GELU}}(x, W, V, W_2) = (\text{GELU}(xW) \otimes xV)W_2$$

$$\text{FFN}_{\text{SwiGLU}}(x, W, V, W_2) = (\text{Swish}_1(xW) \otimes xV)W_2$$

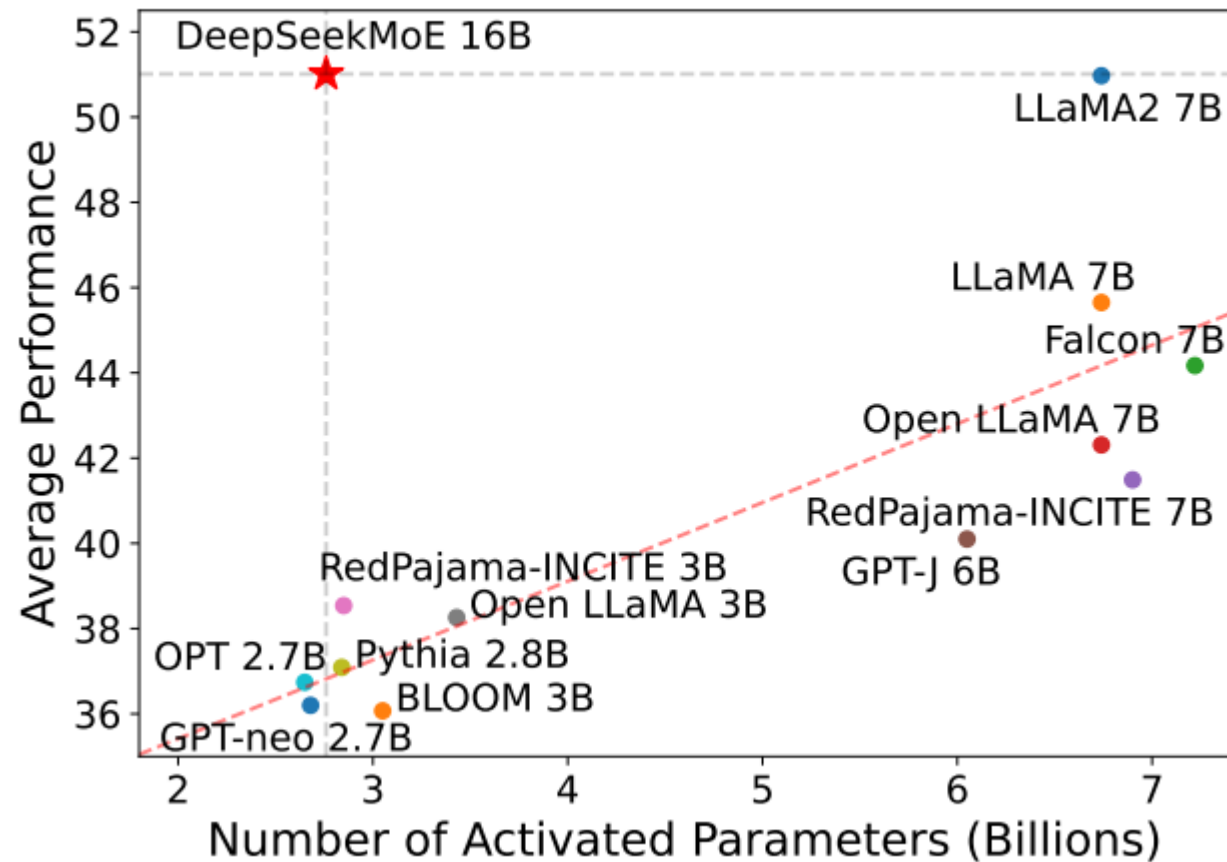
Introducing Mixture of Experts (MoE)



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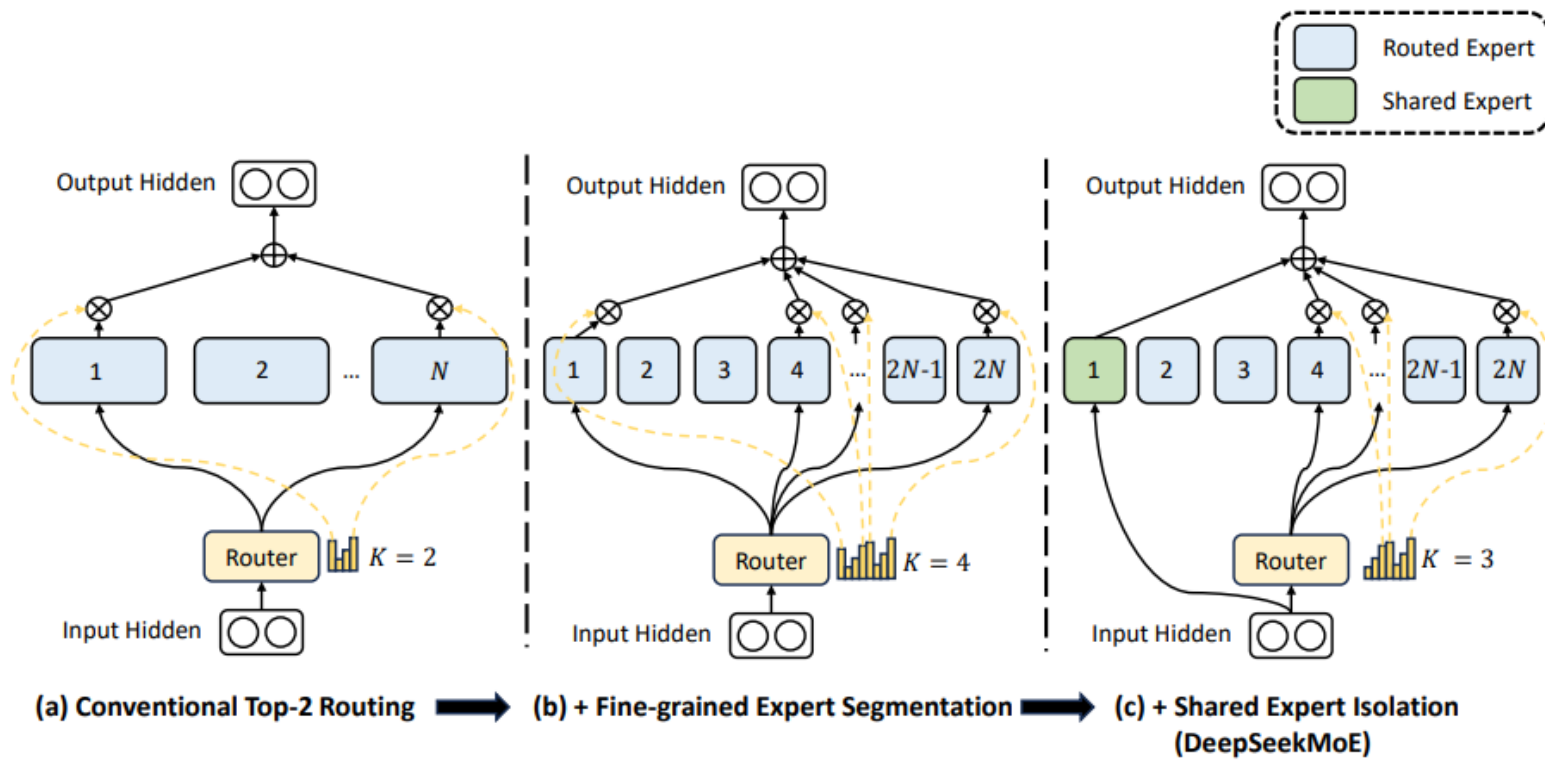
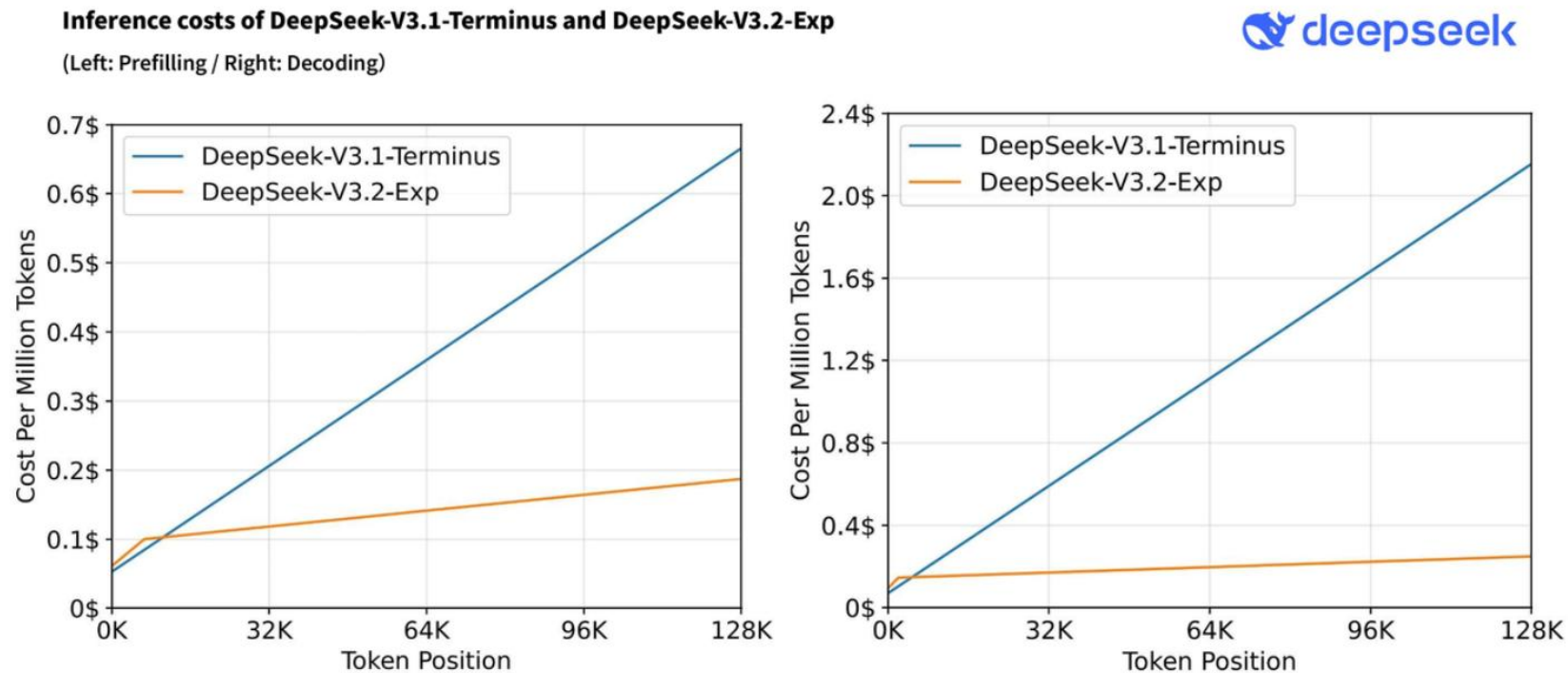




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Introducing Sparse Attention



Introducing Sparse Attention

$$I_{t,s} = \sum_{j=1}^{H^I} w_{t,j}^I \cdot \text{ReLU}(\underbrace{\mathbf{q}_{t,j}^I \cdot \mathbf{k}_s^I}_{\text{dot product}}),$$


$$\mathbf{u}_t = \text{Attn}(\mathbf{h}_t, \{\mathbf{c}_s \mid I_{t,s} \in \text{Top-k}(I_{t,:})\}).$$

Introducing Sparse Attention

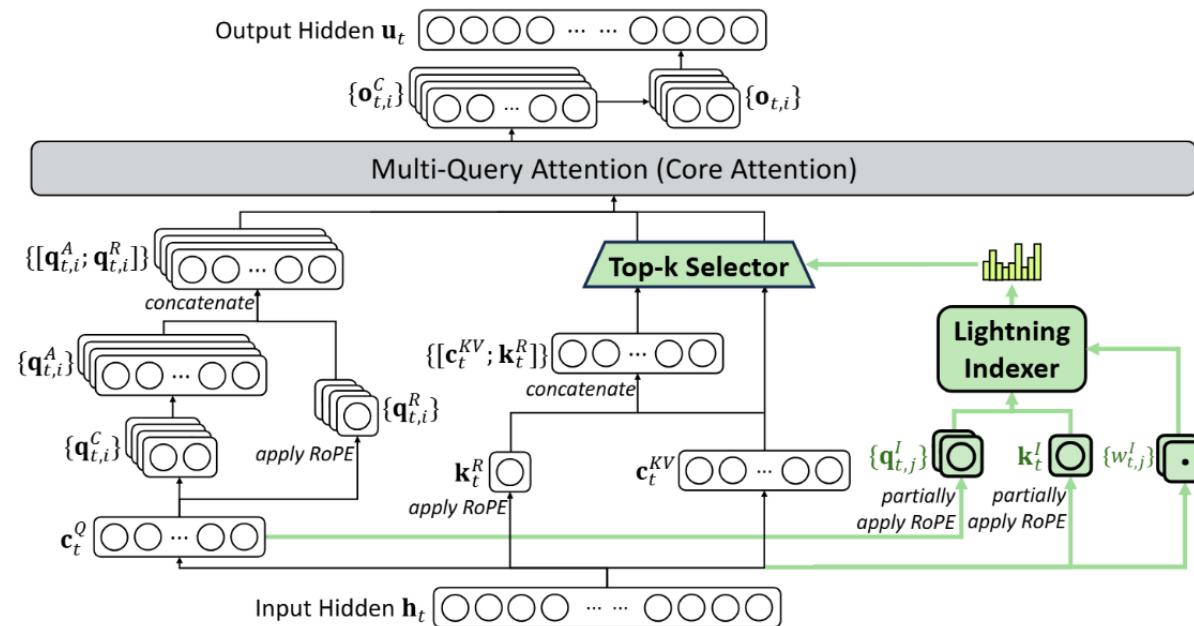


Figure 1 | Attention architecture of DeepSeek-V3.2-Exp, where DSA is instantiated under MLA. The green part illustrates how DSA selects the top-k key-value entries according to the indexer.

